



BobSim

Steady-State Vehicle Characterization

Settled measured- a_y operating points from closed-loop target holds

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Run Notes

- Closed-loop VehicleSim useMode=3 drives each requested target a_y
- Only cases whose steady-state conditions are met are included in the report
- Measured a_y is the analysis axis
- Metrics describe settled endpoints, not ramp-response transients
- $|\text{measured } a_y| < 1.0 \text{ m/s}^2$ is excluded; linear band = 1.0 to 4.0 m/s^2
- Limit metrics use the smooth fitted curves at the first 20% local steering-gradient nonlinearity
- If the fitted 20% threshold is not reached, limit metrics fall back to the largest settled fitted point
- Tire-lift protection terminates if any tire falls below the configured normal-load threshold

SteadyStateEval Summary 12.5 m/s

Metric	Value	Units
Measured a_y Range	1.93 → 18.08	m/s^2
Linear Sideslip Gradient	0.00220	$\frac{\text{rad}}{\text{m/s}^2}$
	1.24	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	−0.00450	$\frac{\text{rad}}{\text{m/s}^2}$
	−2.53	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00060	$\frac{\text{rad}}{\text{m/s}^2}$
	0.34	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	−0.00190	$\frac{\text{rad}}{\text{m/s}^2}$
	−1.07	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.03756	$\frac{\text{rad}}{\text{m/s}^2}$
	21.11	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.02999	$\frac{\text{rad}}{\text{m/s}^2}$
	16.86	$\frac{\text{deg}}{g}$
Linear Roll Gradient	0.883	$\frac{\text{deg}}{g}$
Limit Roll Gradient	0.932	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-20.9	$\text{N} \cdot \text{m}$
Peak Torque a_y	15.99	m/s^2

SteadyStateEval Summary 15.0 m/s

Metric	Value	Units
Measured a_y Range	1.95 → 18.05	m/s^2
Linear Sideslip Gradient	0.00023	$\frac{\text{rad}}{\text{m/s}^2}$
	0.13	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	−0.00757	$\frac{\text{rad}}{\text{m/s}^2}$
	−4.26	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00053	$\frac{\text{rad}}{\text{m/s}^2}$
	0.30	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	−0.00337	$\frac{\text{rad}}{\text{m/s}^2}$
	−1.90	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.02885	$\frac{\text{rad}}{\text{m/s}^2}$
	16.22	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.02471	$\frac{\text{rad}}{\text{m/s}^2}$
	13.89	$\frac{\text{deg}}{g}$
Linear Roll Gradient	0.887	$\frac{\text{deg}}{g}$
Limit Roll Gradient	0.969	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-21.0	$\text{N} \cdot \text{m}$
Peak Torque a_y	18.05	m/s^2

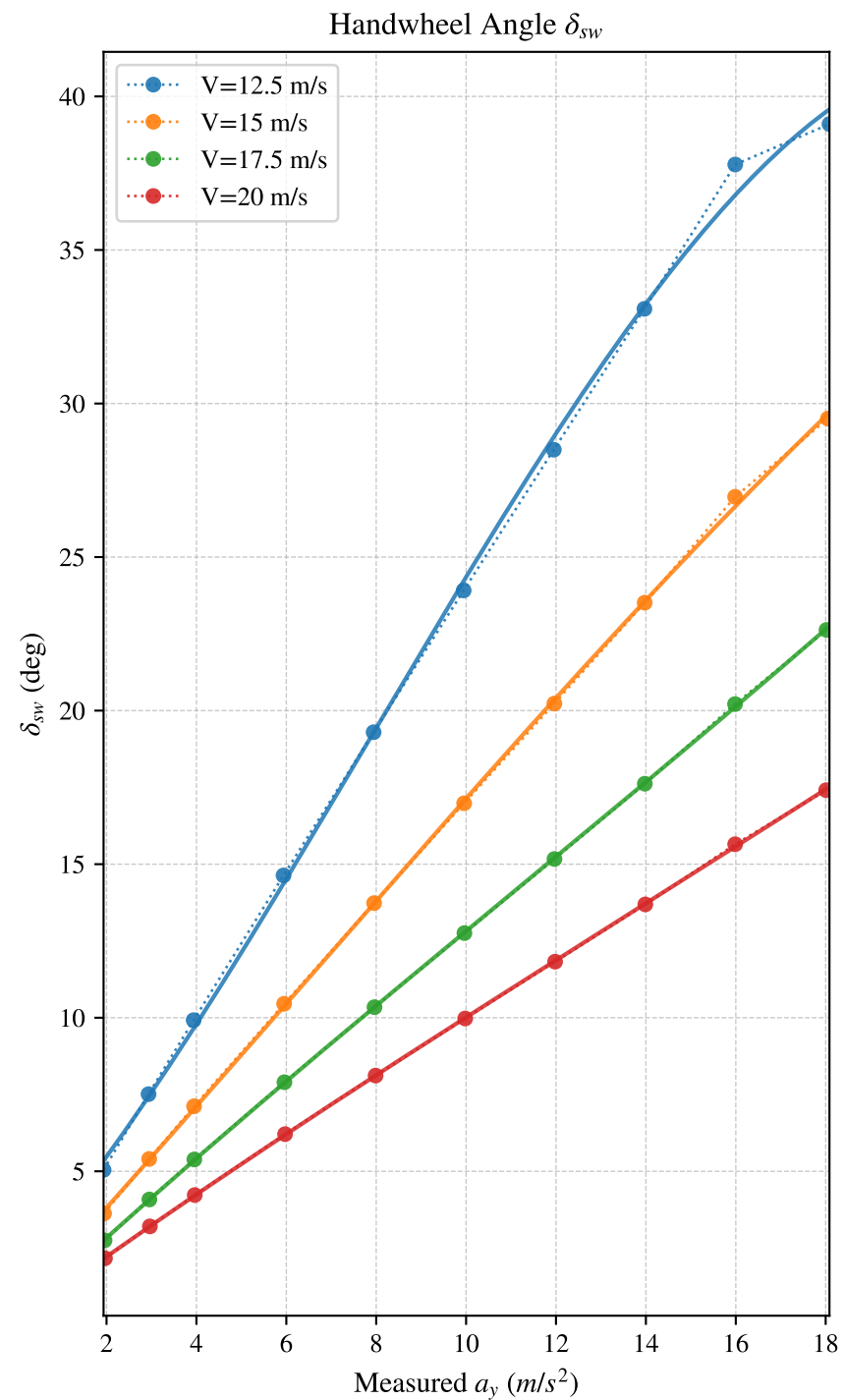
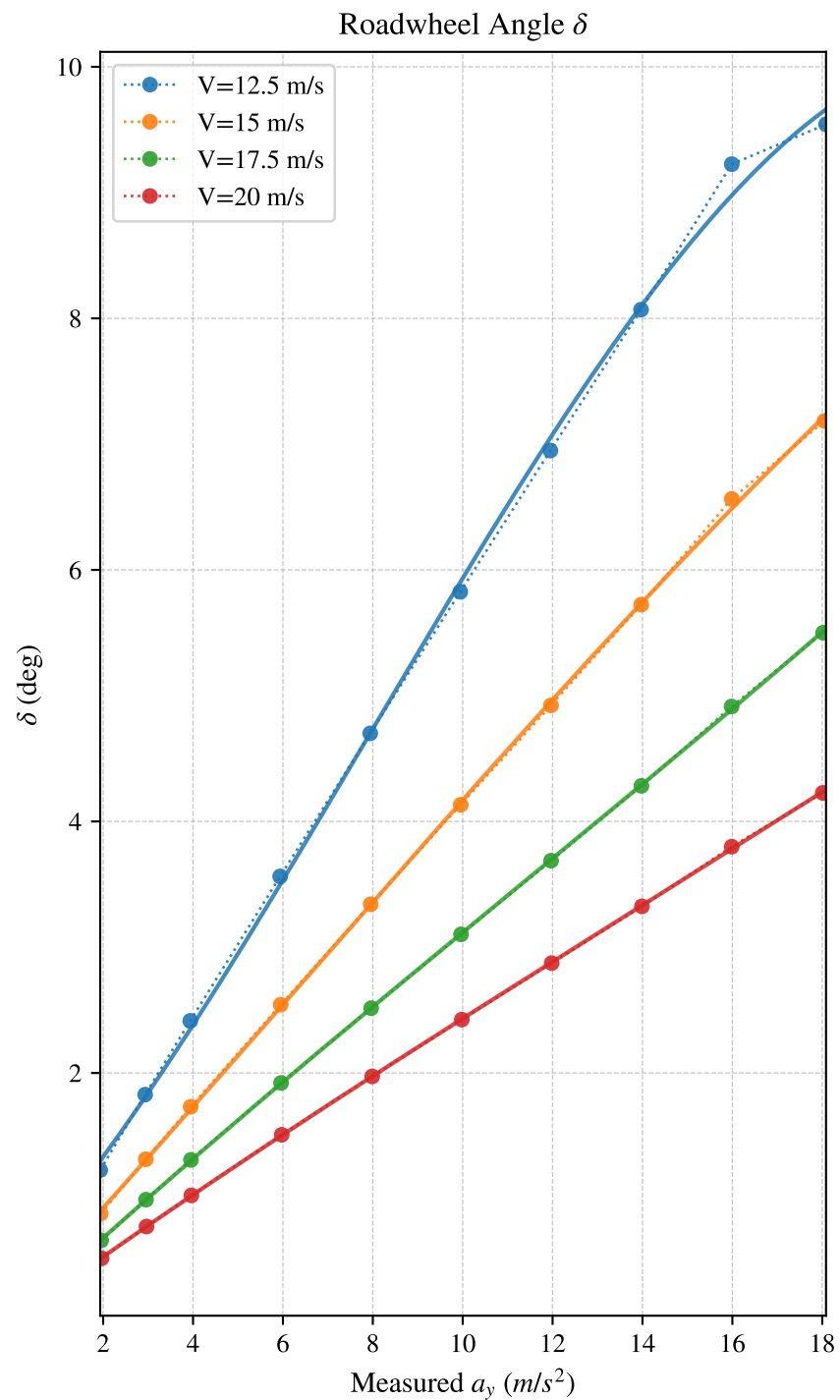
SteadyStateEval Summary 17.5 m/s

Metric	Value	Units
Measured a_y Range	1.96 → 18.02	m/s^2
Linear Sideslip Gradient	−0.00027	$\frac{\text{rad}}{\text{m/s}^2}$
	−0.15	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	−0.00649	$\frac{\text{rad}}{\text{m/s}^2}$
	−3.65	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00053	$\frac{\text{rad}}{\text{m/s}^2}$
	0.30	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	−0.00017	$\frac{\text{rad}}{\text{m/s}^2}$
	−0.10	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.02257	$\frac{\text{rad}}{\text{m/s}^2}$
	12.69	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.02223	$\frac{\text{rad}}{\text{m/s}^2}$
	12.50	$\frac{\text{deg}}{g}$
Linear Roll Gradient	0.893	$\frac{\text{deg}}{g}$
Limit Roll Gradient	0.980	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-21.5	$\text{N} \cdot \text{m}$
Peak Torque a_y	18.02	m/s^2

SteadyStateEval Summary 20.0 m/s

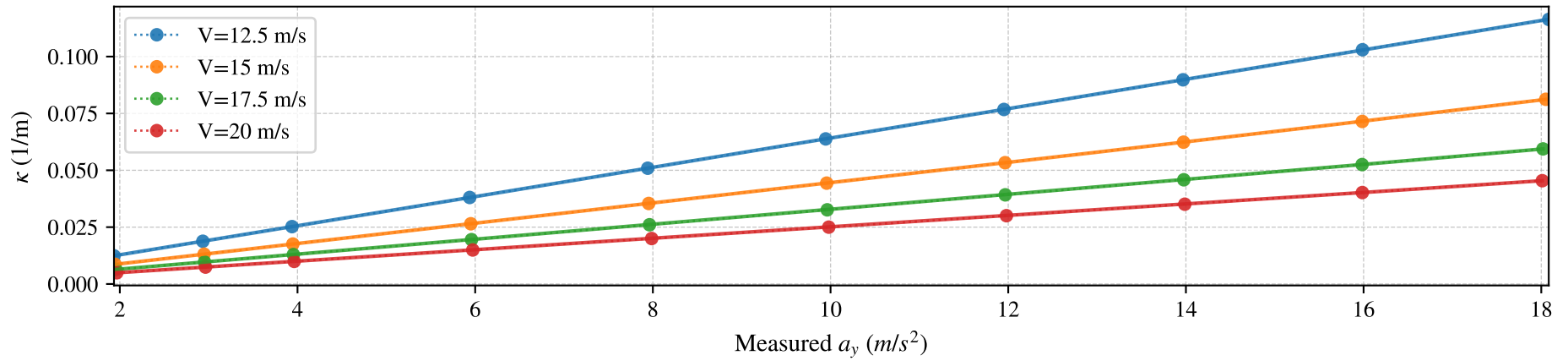
Metric	Value	Units
Measured a_y Range	1.97 → 18.01	m/s^2
Linear Sideslip Gradient	-0.00067	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.38	$\frac{\text{deg}}{g}$
Limit Sideslip Gradient	-0.00616	$\frac{\text{rad}}{\text{m/s}^2}$
	-3.46	$\frac{\text{deg}}{g}$
Linear Steer-Excess Understeer Gradient	0.00050	$\frac{\text{rad}}{\text{m/s}^2}$
	0.28	$\frac{\text{deg}}{g}$
Limit Steer-Excess Understeer Gradient	-0.00018	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.10	$\frac{\text{deg}}{g}$
Linear Raw Handwheel Angle Gradient	0.01769	$\frac{\text{rad}}{\text{m/s}^2}$
	9.94	$\frac{\text{deg}}{g}$
Limit Raw Handwheel Angle Gradient	0.01637	$\frac{\text{rad}}{\text{m/s}^2}$
	9.20	$\frac{\text{deg}}{g}$
Linear Roll Gradient	0.899	$\frac{\text{deg}}{g}$
Limit Roll Gradient	0.991	$\frac{\text{deg}}{g}$
Peak Handwheel Torque	-22.0	$\text{N} \cdot \text{m}$
Peak Torque a_y	18.01	m/s^2

Steady-State Steering Points vs Measured a_y

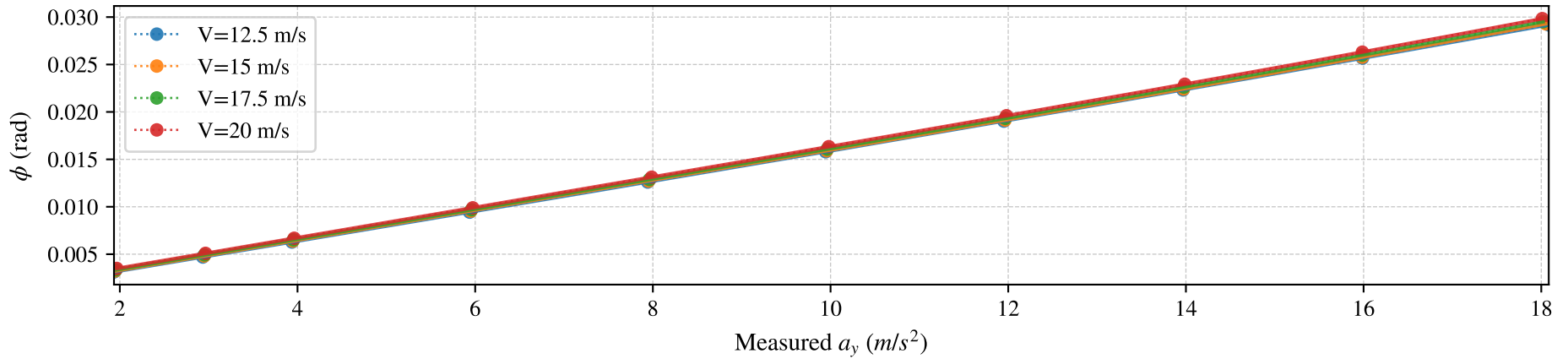


Steady-State Vehicle States vs Measured a_y

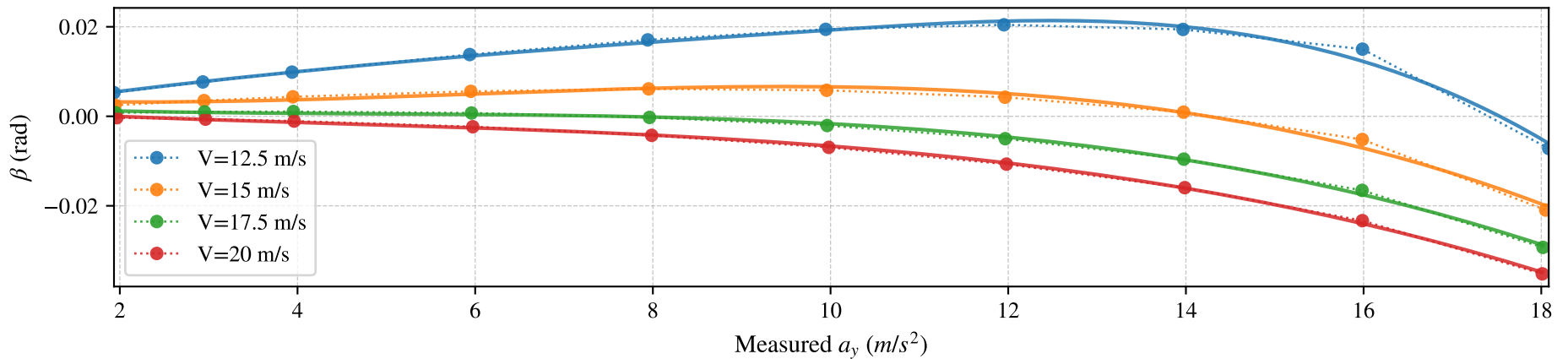
Curvature κ



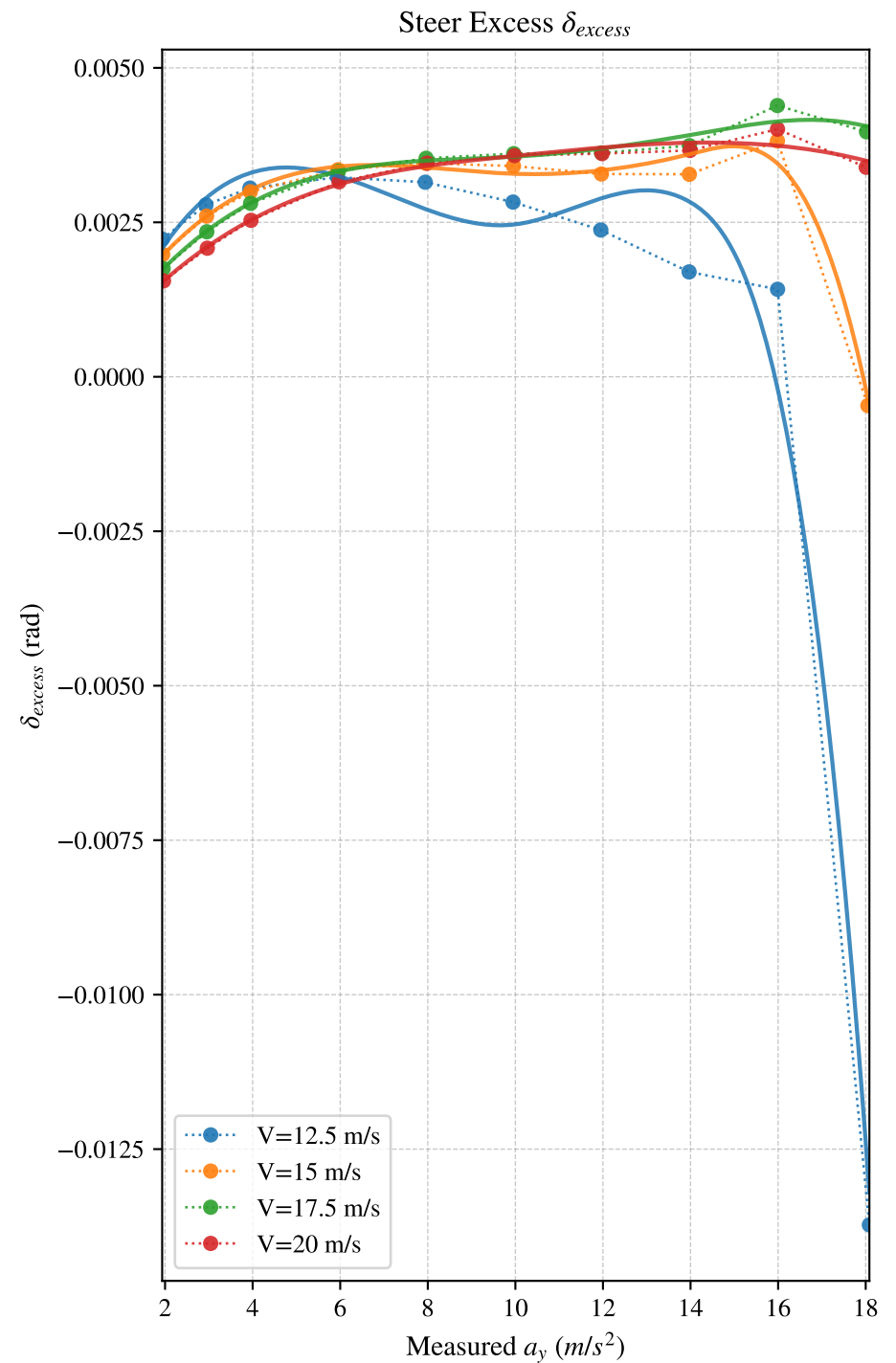
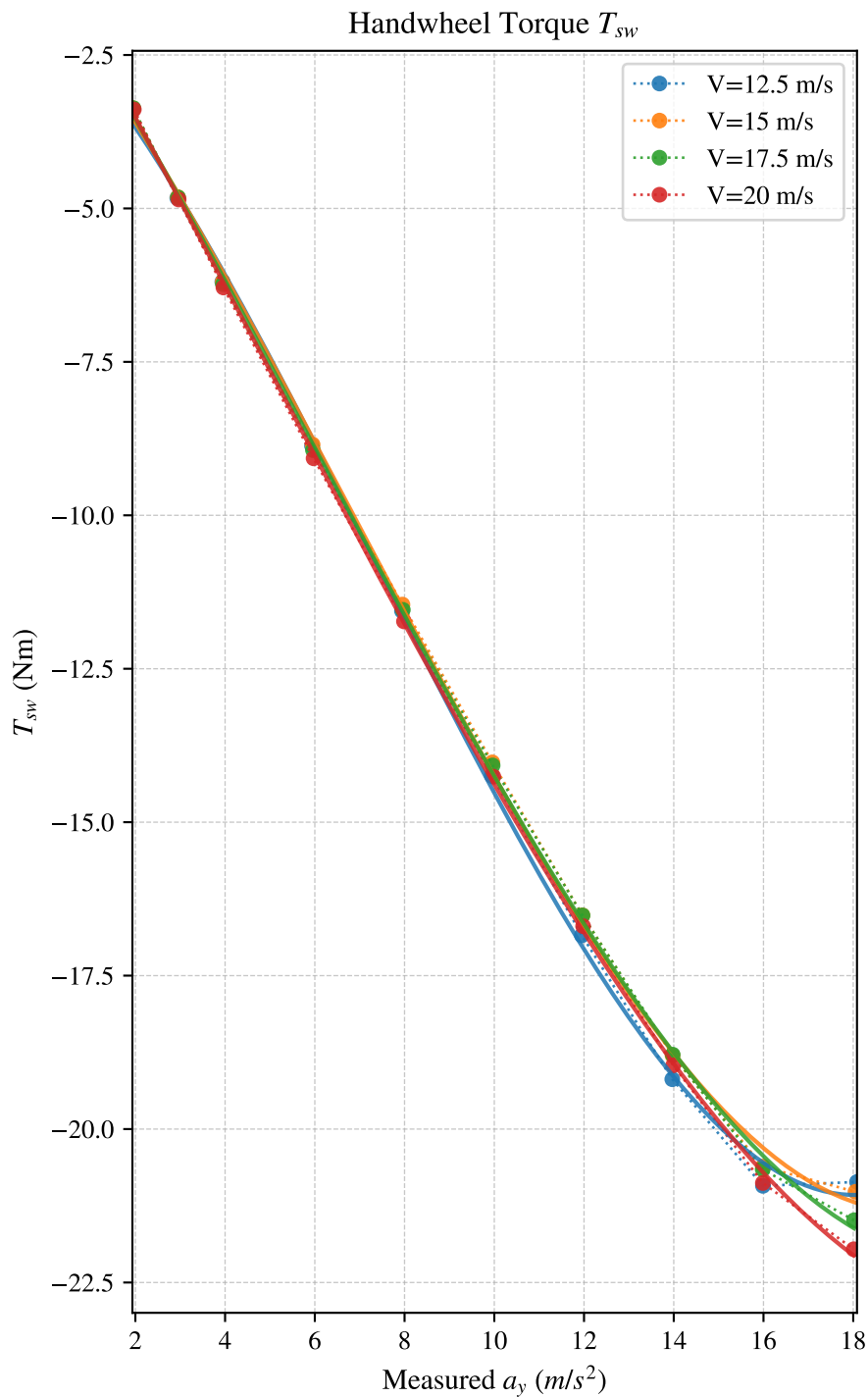
Roll Angle ϕ



Sideslip β



Steady-State Steering Effort vs Measured a_y



Steady-State Control Effort vs Measured a_y

